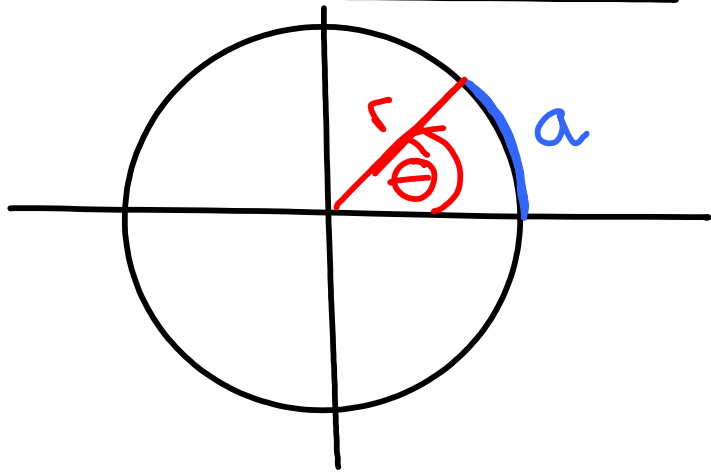


Radiaalmaat en Trigonometrische Funksies

(Studiegids
laaste bl.)

Definisie:



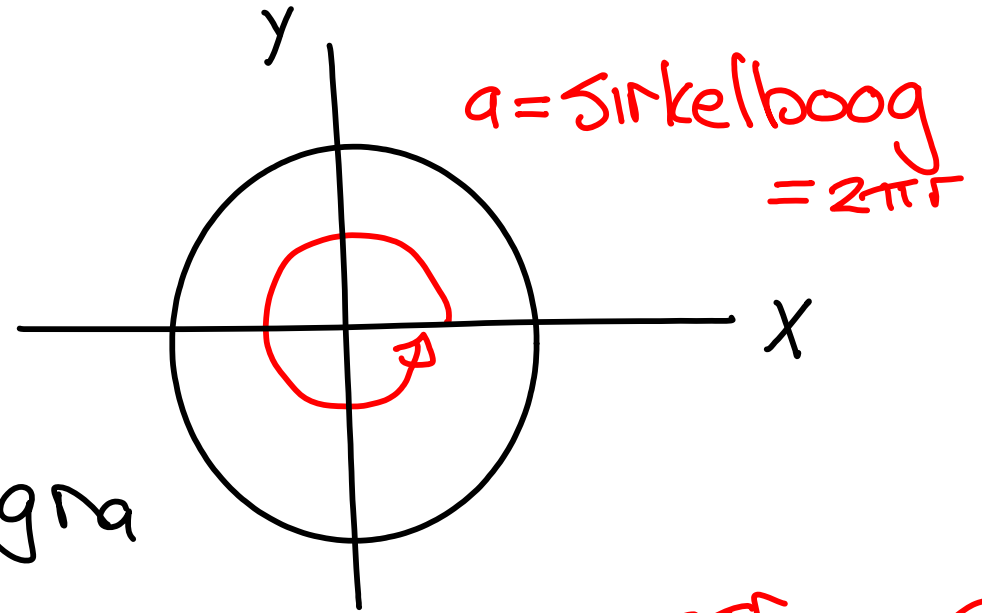
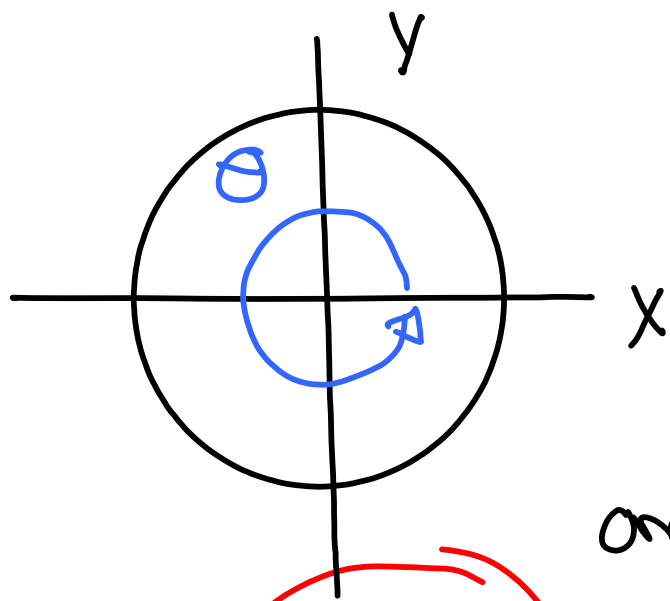
$$\pi \text{ radiale} = 180^\circ$$

a is 'n. sirkelboog

$$\theta = \frac{a}{r} \text{ radiale}$$

$$\pi = \text{pi} \approx 3.14 \text{ (reële getalle)}$$

Omskakeling tussen grade en meting in radiale.



$$a = \text{Sirkelboog} = 2\pi r$$

• $\Theta = 360^\circ$

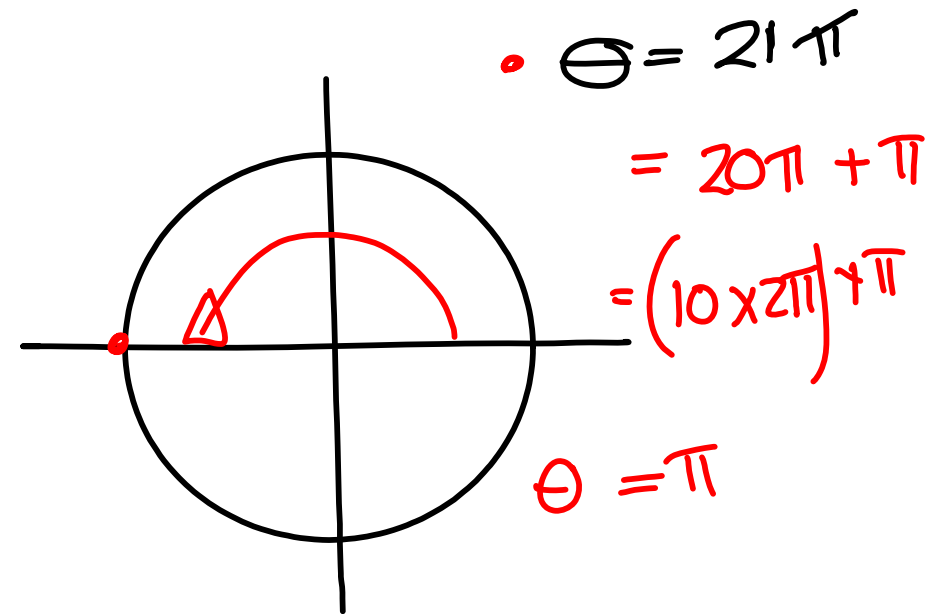
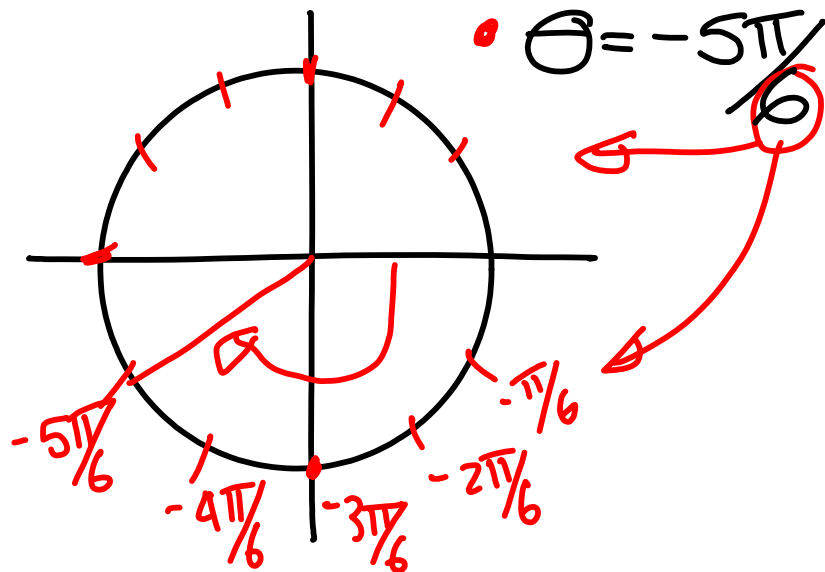
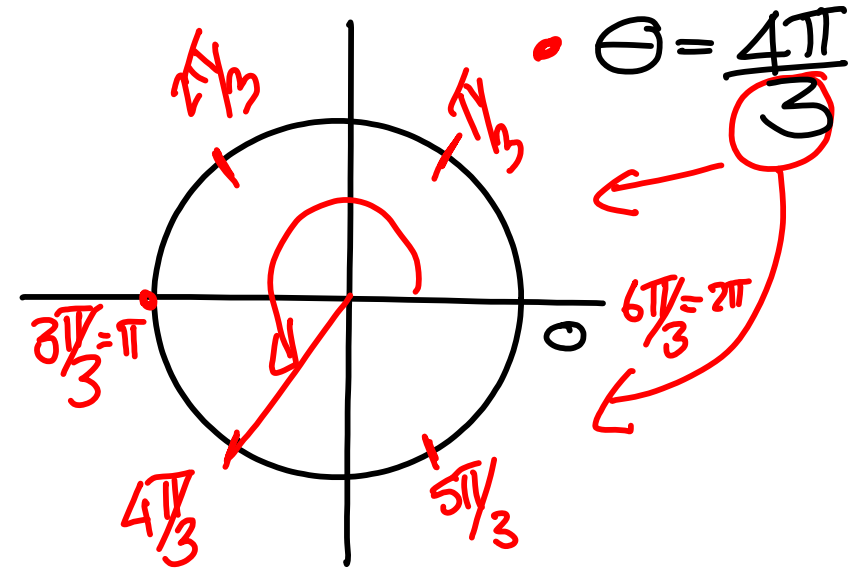
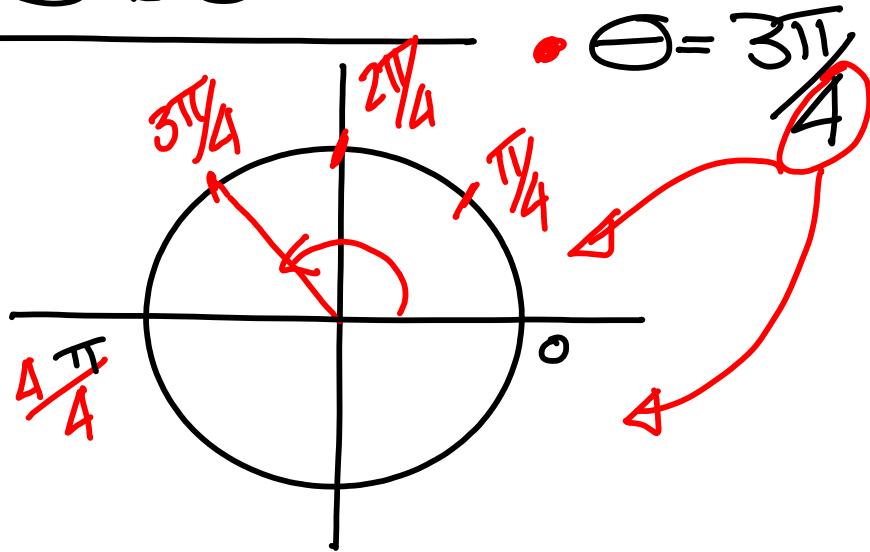
omskabeling na
radiale

$$\Theta = \frac{a}{r} = \frac{2\pi r}{r} = 2\pi$$

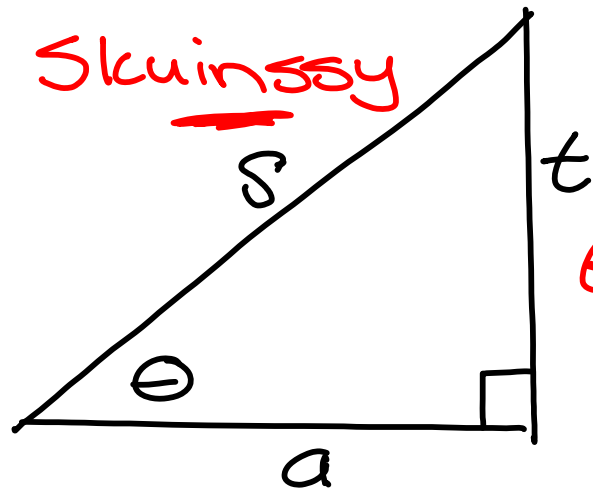
360° is 2π radiale
 180° is π radiale

Grade	0°	30°	45°	60°	90°	180°	360°
Radiale	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π

Sketchs:



Definisie : 6 Trigonometrisiese Funksies



teenoorstaande sy

aangrensende sy

$$\bullet \sin \theta = \frac{t}{s}$$

$$\bullet \cos \theta = \frac{a}{s}$$

$$\bullet \tan \theta = \frac{t}{a}$$

$$\bullet \csc \theta = \frac{s}{t}$$

$$\bullet \sec \theta = \frac{s}{a}$$

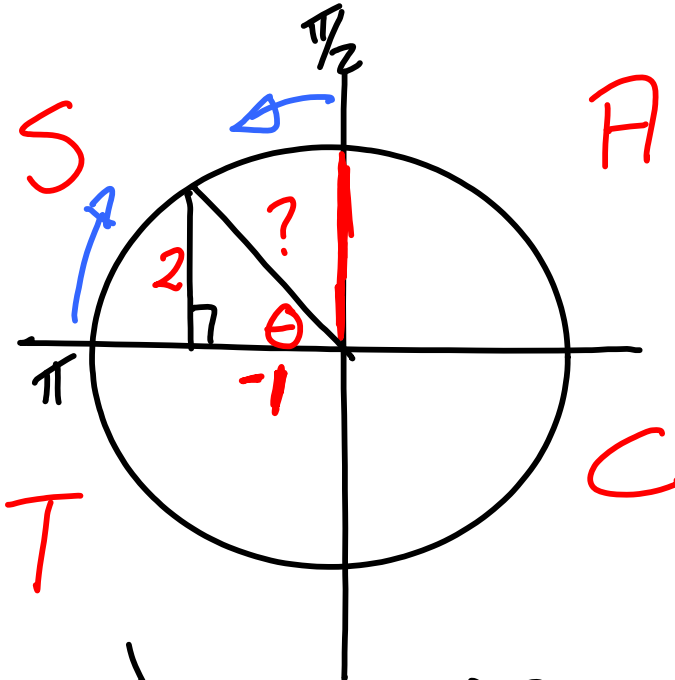
$$\bullet \cot \theta = \frac{a}{t}$$

Tekens van Trig funkties

Pythagoras:

$$2^2 + (-1)^2 = 5$$

$$\text{Hypotenys} = \sqrt{5}$$



Gegee:

$$\bullet \tan \theta = \frac{-2}{1}$$

met

$$\frac{\pi}{2} < \theta < \pi$$

- Vind waarden van ander trigonometrische verhoudings.

$$\sin \theta = \frac{1}{\sqrt{5}}$$

$$\operatorname{cosec} \theta = \frac{\sqrt{5}}{1}$$

$$\cot \theta = -\frac{1}{2}$$

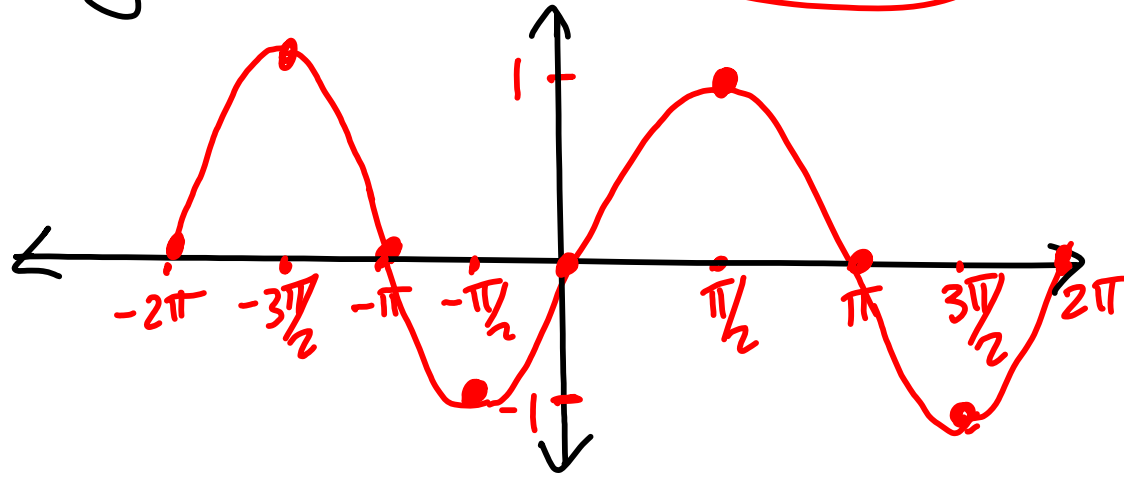
$$\cos \theta = -\frac{2}{\sqrt{5}}$$

$$\sec \theta = -\frac{\sqrt{5}}{2}$$

Grafieke

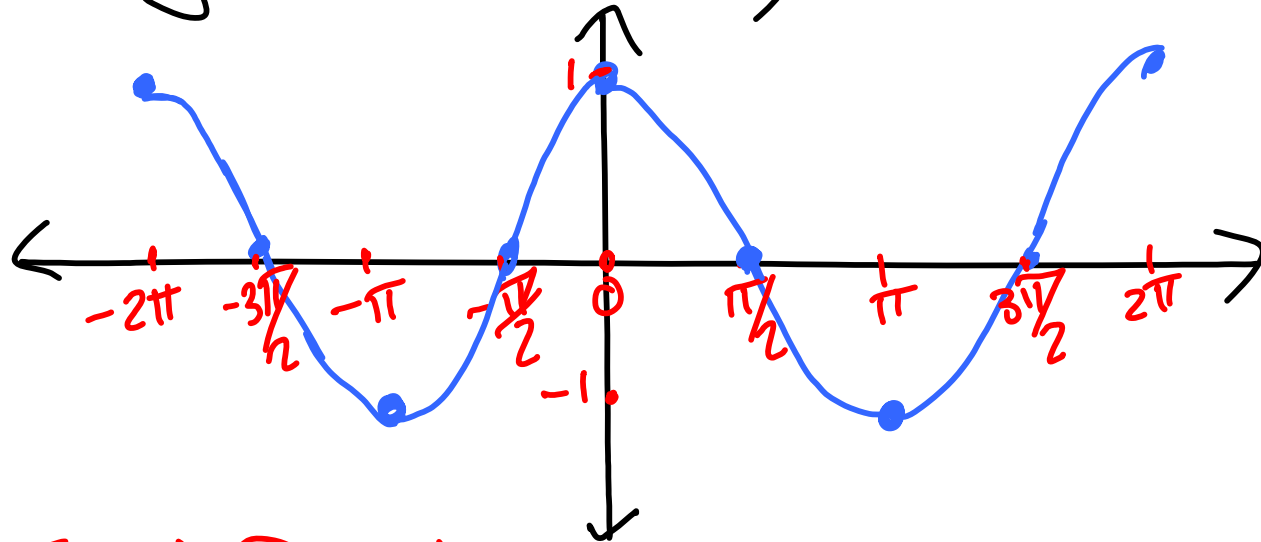
A 30 (Moet kan skets)

• $y = \sin x, x \in \mathbb{R}$



• $D_f = \mathbb{R}$
 $W_f = \{y \mid -1 \leq y \leq 1\}$
 $= [-1, 1]$

• $y = \cos x, x \in \mathbb{R}$



$D_f = \mathbb{R}$
 $W_f = [-1, 1]$

Selfstudie : $y = \tan x$

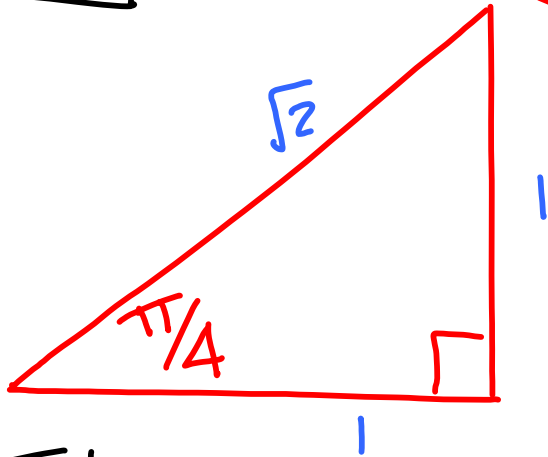
kyk!!! teken!!
Doen dit ✓

Bepaal trigonometrische Verhoudings sonder sakrekenaar.

$\frac{\pi}{4}$

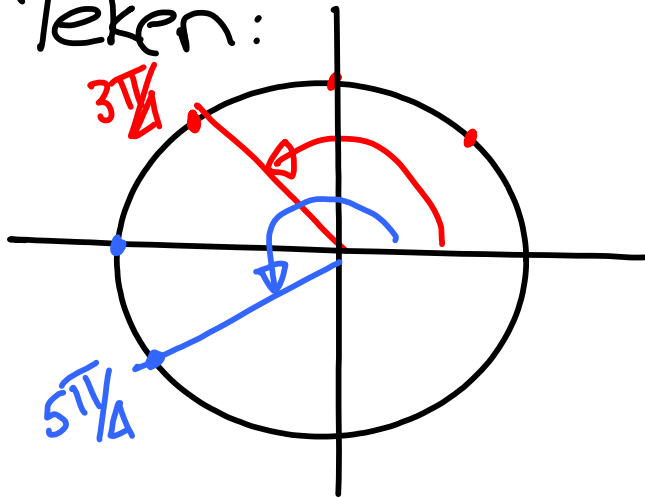
Verwysingshoek

$$\left(\begin{array}{l} \operatorname{cosec} \frac{\pi}{4} = \sqrt{2} \\ \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}} \end{array} \right)$$



$$\bullet \operatorname{cosec} \frac{3\pi}{4} = + \frac{\sqrt{2}}{1} = \sqrt{2}$$

• Teken:

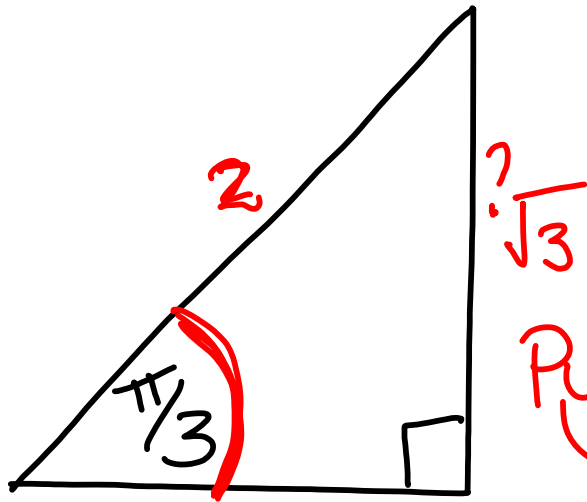


$$\bullet \cos \frac{5\pi}{4} = -\frac{1}{\sqrt{2}}$$

$$\boxed{\frac{\pi}{3}}$$

verwysingshoek

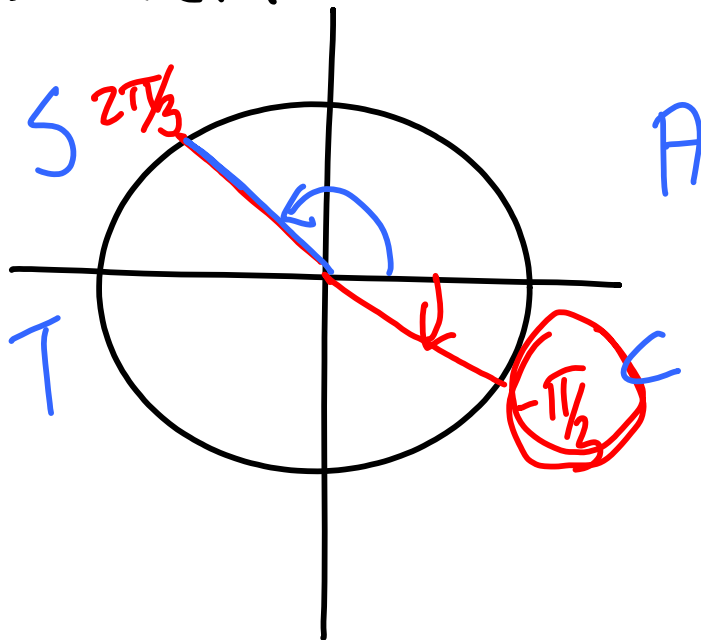
$$\left(\begin{array}{l} \bullet \tan \frac{\pi}{3} = \frac{\sqrt{3}}{1} \\ \bullet \sec \frac{\pi}{3} = \frac{2}{1} \end{array} \right)$$



Pythagoras

$$\bullet \tan \left(-\frac{\pi}{3} \right) = -\frac{\sqrt{3}}{1}$$

• Teken:

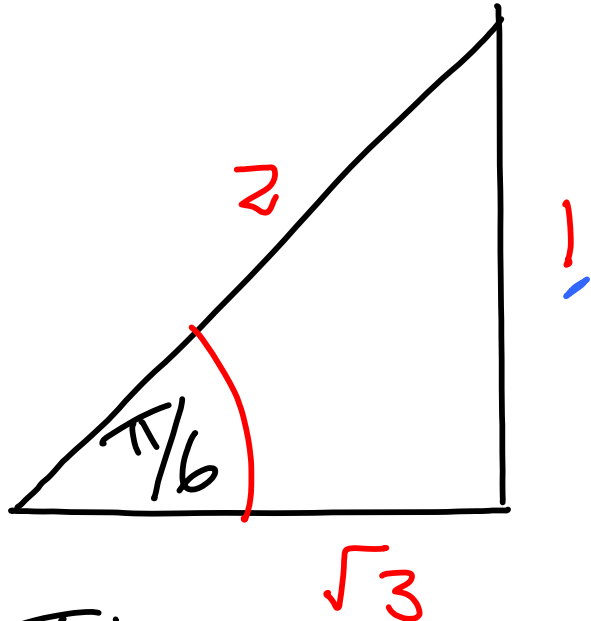


$$\bullet \sec \frac{2\pi}{3} = -2$$

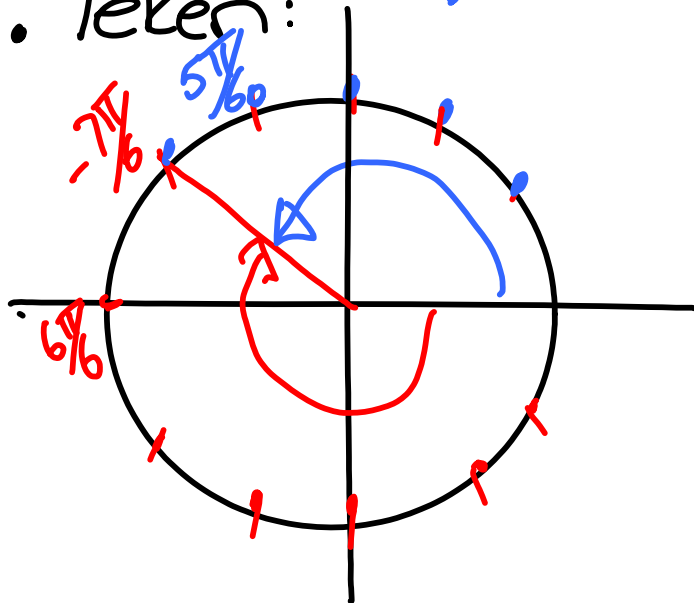
Oefen die trig
werkkartaar op
ClickUp Deel A

$$\boxed{\frac{\pi}{6}}$$

Verwysingshoek



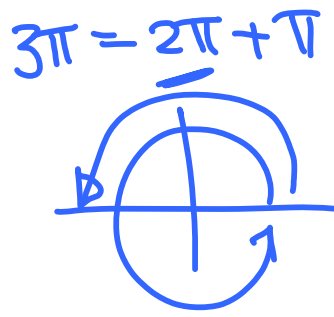
• Tekening:



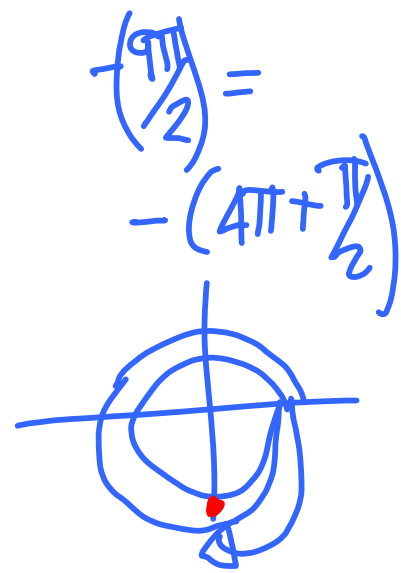
$$\bullet \sin\left(\frac{-\pi}{6}\right) = +\frac{1}{2}$$

$$\bullet \cot\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{1} = -\sqrt{3}$$

- $\sin(3\pi)$
 $= \sin \pi$
 $= 0$



- $\cos(-\frac{9\pi}{2})$
 $= \cos(-\frac{\pi}{2})$
 $= 0$

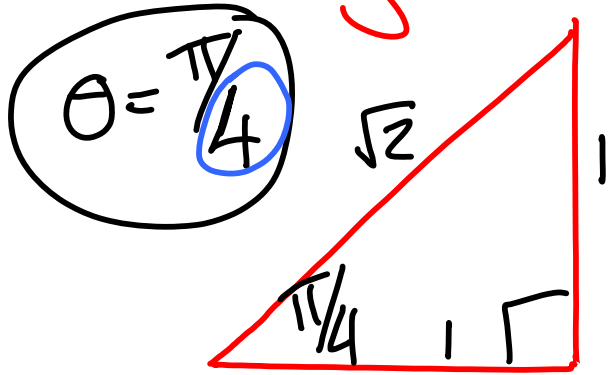


Oplos van Trigonometrische

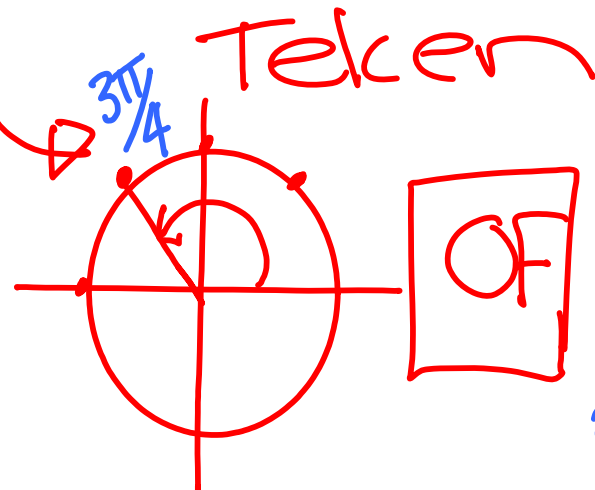
Vergelykings

① $\cos(2x) = \boxed{-}\frac{1}{\sqrt{2}}$

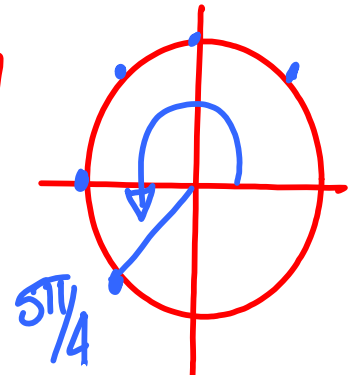
Verwysingshoek



(geen beperking)



OF



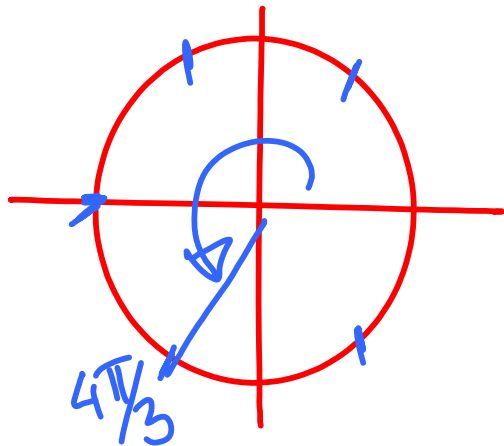
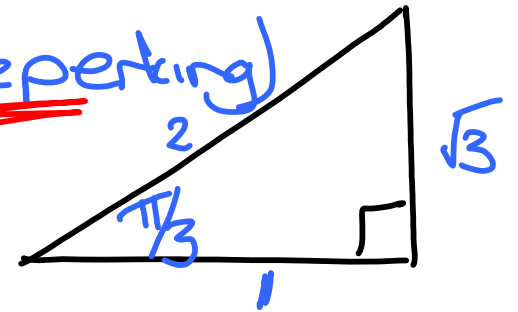
$$2x = \frac{3\pi}{4} + 2k\pi \quad \text{of} \quad 2x = \frac{5\pi}{4} + 2k\pi$$

$$\Rightarrow x = \frac{3\pi}{8} + k\pi \quad \text{of} \quad x = \frac{5\pi}{8} + k\pi$$

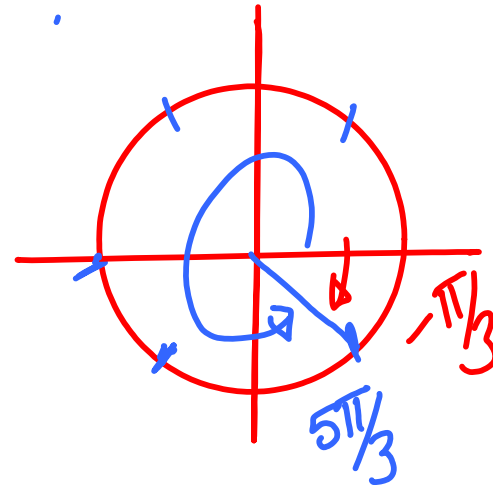
en $k \in \mathbb{Z}$

② $\csc x = -\frac{2}{\sqrt{3}}$ (geen beperking)

- Verwijshoek: $\frac{\pi}{3}$
- Tekens:



of



$$x = \frac{4\pi}{3} + 2k\pi \quad \text{of} \quad x = \frac{5\pi}{3} + 2k\pi \quad // \quad x = -\frac{\pi}{3} + 2k\pi$$

$k \in \mathbb{Z}$

Maar wat van??

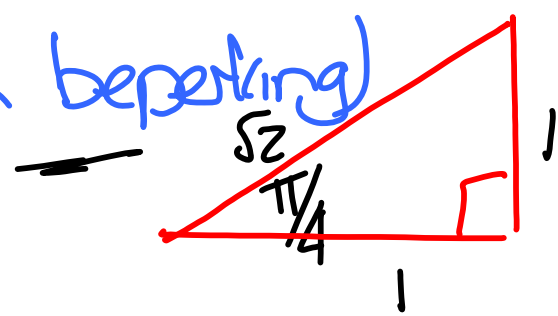
$$\operatorname{cosec} x = -\frac{2}{\sqrt{3}}, \quad \text{Bepending!!}$$

$x \in [0, 2\pi]$

Dan is $x = \frac{4\pi}{3}$ of $x = \frac{5\pi}{3}$

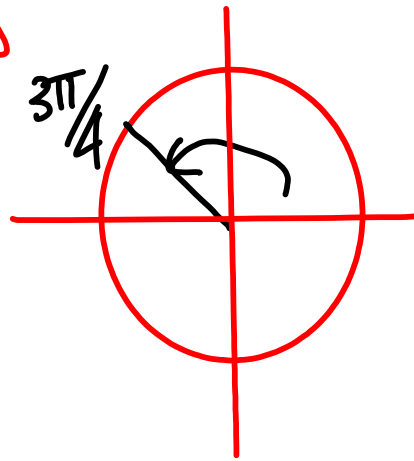
Wees versigtig: $-\frac{\pi}{3} \notin [0, 2\pi]$

③ $\tan \frac{x}{2} = -1$ (Geen beperking)

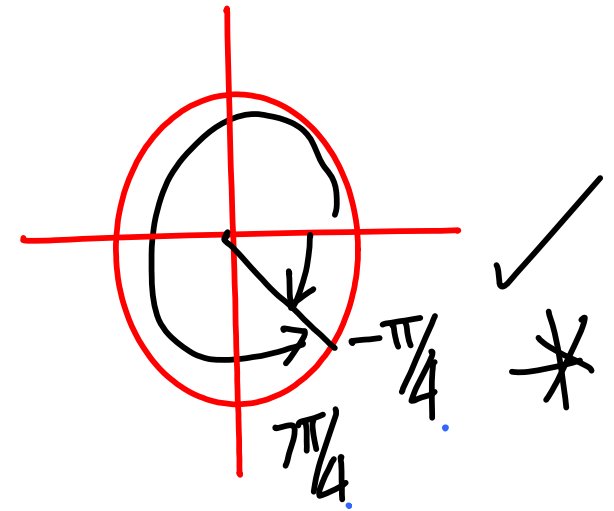


- Verwysingshoek: $\frac{\pi}{4}$

• Teken



of



$$\frac{x}{2} = \frac{3\pi}{4} + k\pi \quad \text{of} \quad \frac{x}{2} = -\frac{\pi}{4} + k\pi$$

$$x = \frac{3\pi}{2} + 2k\pi \quad \text{of} \quad x = -\frac{\pi}{2} + 2k\pi$$

en $k \in \mathbb{Z}$

*ken Identiteite : A28

④ $\sin 2x = \cos x$ op $[0, 2\pi]$ *Bepending!*

$\Rightarrow 2 \sin x \cos x = \cos x$

$\Rightarrow 2 \sin x \cos x - \cos x = 0$

$\Rightarrow \cos x (2 \sin x - 1) = 0$

$\Rightarrow \cos x = 0$ of $2 \sin x - 1 = 0$

$\Rightarrow \cos x = 0$ of $\sin x = +\frac{1}{2}$

$\Rightarrow x = \frac{\pi}{2}$ of $x = \frac{3\pi}{2}$ of $x = \frac{\pi}{6}$ of $x = \frac{5\pi}{6}$

Gave reël
moenie uitdeelnie

Bepending
 $[0, 2\pi]$

$$⑤ \quad 2 \cos^2 x - \sin x - 1 = 0$$

$$\Rightarrow 2(1 - \sin^2 x) - \sin x - 1 = 0$$

$$\Rightarrow 2 - 2\sin^2 x - \sin x - 1 = 0$$

$$\Rightarrow -2\sin^2 x - \sin x + 1 = 0$$

$$\stackrel{x(-1)}{\Rightarrow} 2\sin^2 x + \sin x - 1 = 0$$

$$\Rightarrow (2\sin x - 1)(\sin x + 1) = 0$$

$$\Rightarrow 2\sin x - 1 = 0 \quad \text{of} \quad \sin x + 1 = 0$$

$$\Rightarrow \sin x = +\frac{1}{2} \quad \text{of} \quad \sin x = -1$$

$$\Rightarrow x = \frac{\pi}{6} + 2k\pi \quad \text{of} \quad x = \frac{5\pi}{6} + 2k\pi \quad \text{of} \quad x = \frac{3\pi}{2} + 2k\pi$$

Geen
Bepertking
(+2kπ)

Identiteit
 $\sin^2 x + \cos^2 x = 1$

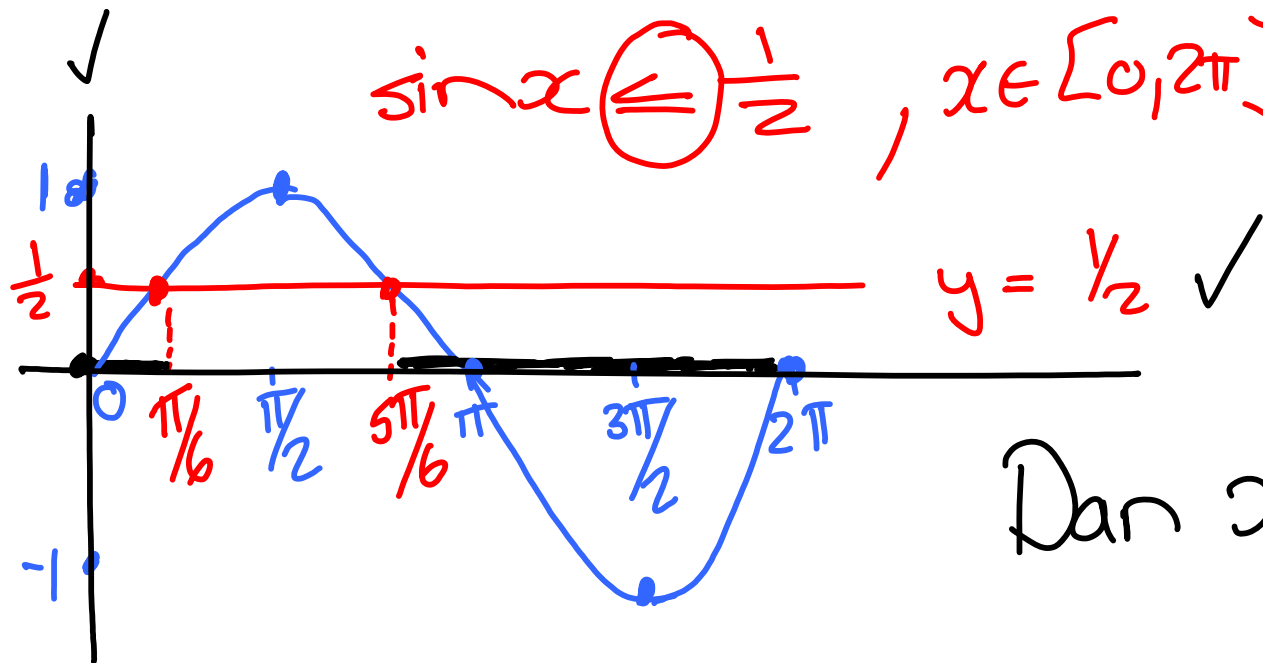
(Doen Triguwerkkraat B)

$k \in \mathbb{Z}$

Trig Onge lykhede

* Moet grafiek gebruik

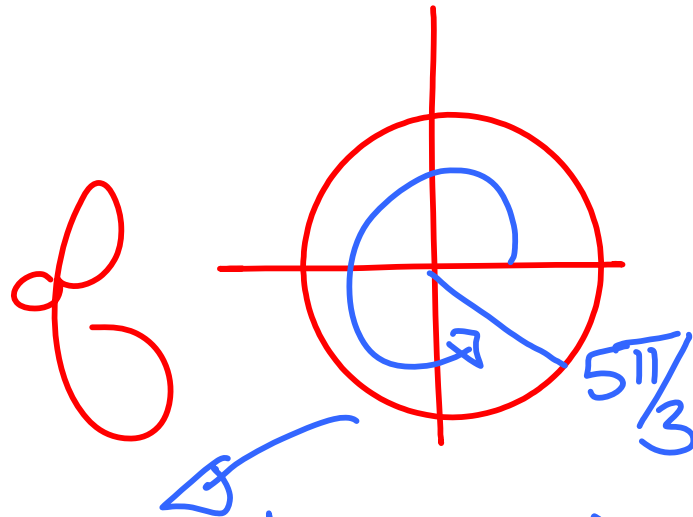
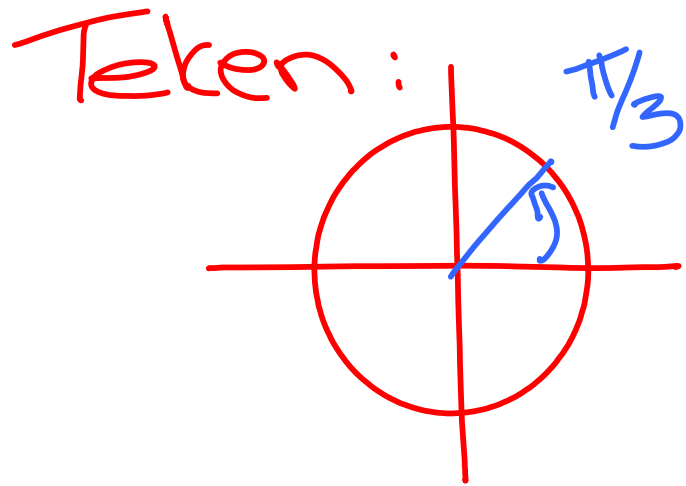
① $\sin x \leq \frac{1}{2}$, $x \in [0, 2\pi]$
 $\sin x = +\frac{1}{2}$
 $\Rightarrow x = \frac{\pi}{6}$ of $x = \frac{5\pi}{6}$
Verwijzing $\frac{\pi}{6}$ is $\frac{\pi}{6}$ ✓



✓

Dan $x \in [0, \frac{\pi}{6}] \cup [\frac{5\pi}{6}, 2\pi]$

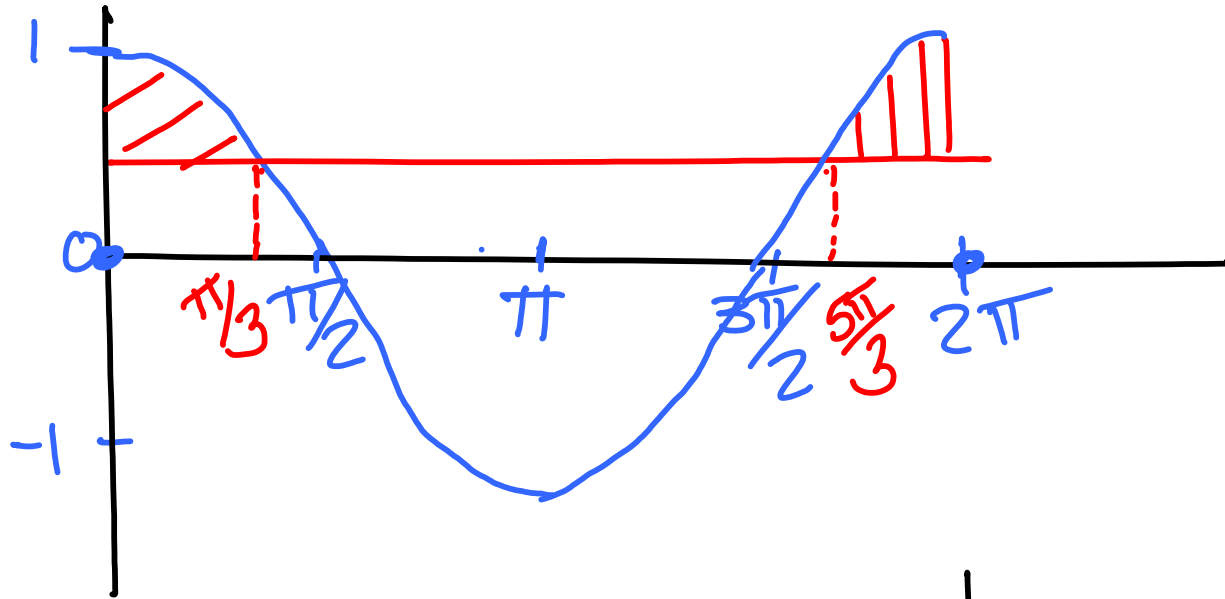
② $2\cos x - 1 > 0$, $x \in [0, 2\pi]$
 $\cos x = +\frac{1}{2}$ Verwijshoek : $\frac{\pi}{3}$



Onthou nou : ons kan nie $-\frac{\pi}{3}$
 gebruik nie want $-\frac{\pi}{3} \notin [0, 2\pi]$
 maar $\frac{5\pi}{3} \in [0, 2\pi]$

Grafiek : $\cos x > \frac{1}{2}$, $x \in [0, 2\pi]$

Onthou om die gestakeerde gebied
op die grafiek aan te toon
of donker lyn op x-as



Dan is $\cos x > \frac{1}{2}$ as
 $x \in [0, \pi/3) \cup (5\pi/3, 2\pi]$

③ $\tan x > -1$ op $[-\pi, \pi]$ ←

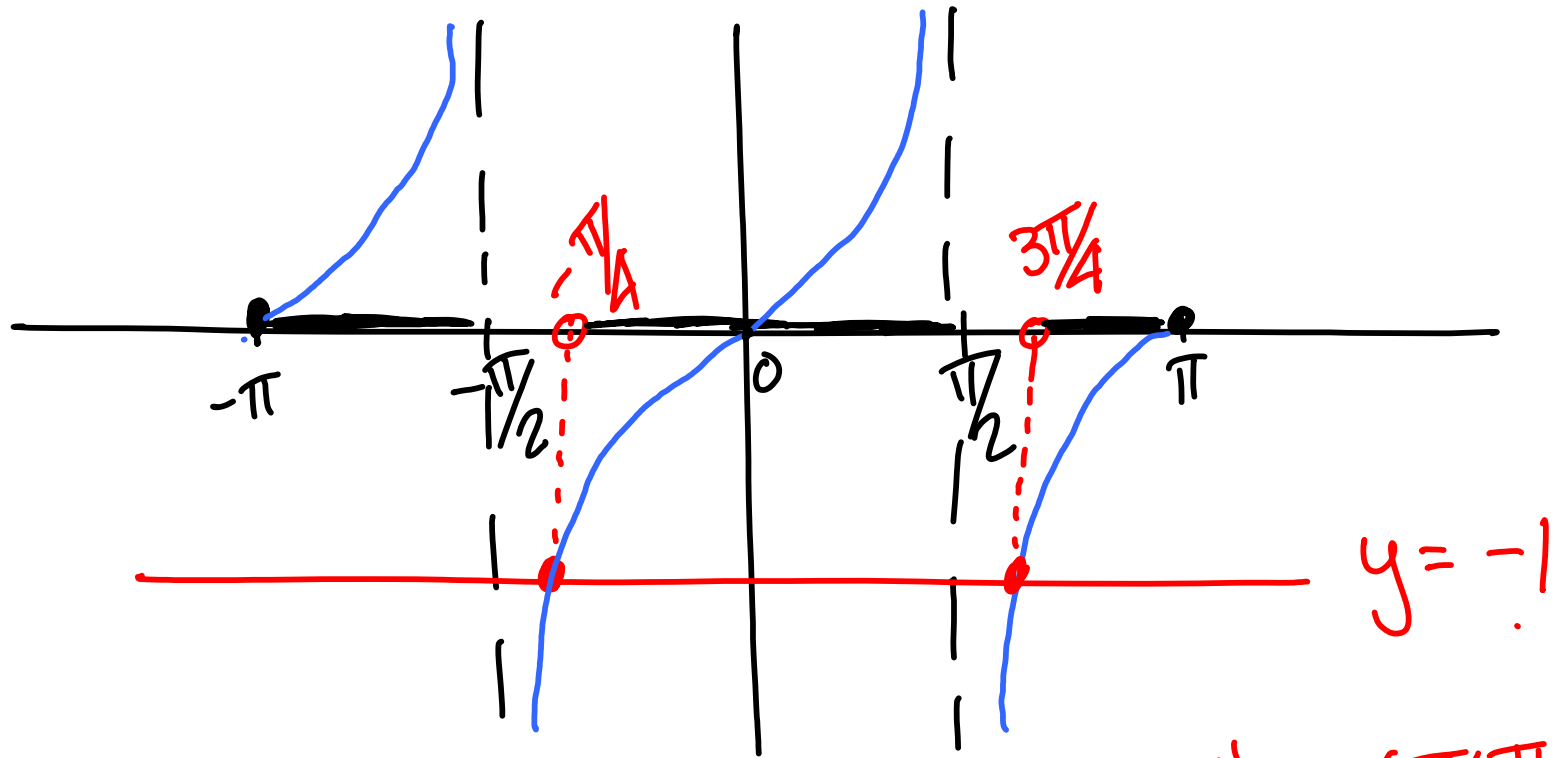
$\tan x = -1$

Verwijnghoek: $x = \frac{\pi}{4}$

Teken $\frac{3\pi}{4}$

$x = \frac{3\pi}{4}$ of $x = -\frac{\pi}{4}$

? $-\frac{\pi}{4}$ of $\frac{\pi}{4}$



Dus $\tan x > -1$ op $x \in [-\pi, -\frac{\pi}{2}) \cup (-\frac{\pi}{4}, \frac{\pi}{2}) \cup (\frac{3\pi}{4}, \pi]$

Probeer die volgende

1. $|\sin x| \leq \frac{1}{2}$, $x \in [0, 2\pi]$
2. $\cos x > \sin x$, $x \in [0, 2\pi]$
3. $|\cos x| \geq \frac{1}{2}$, $x \in [0, 2\pi]$
4. $0 \leq \tan x \leq 1$, $x \in [0, 2\pi]$

Antwoorden op Clickup onder
Extra trig problemen